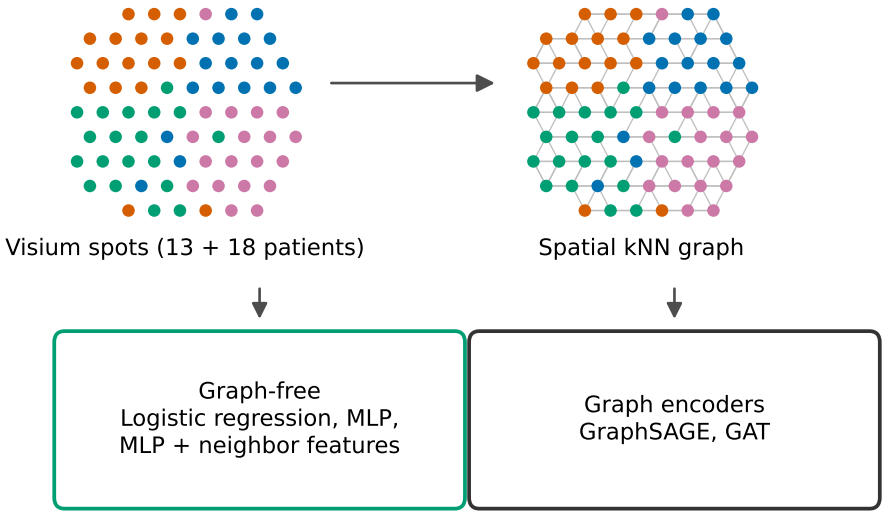
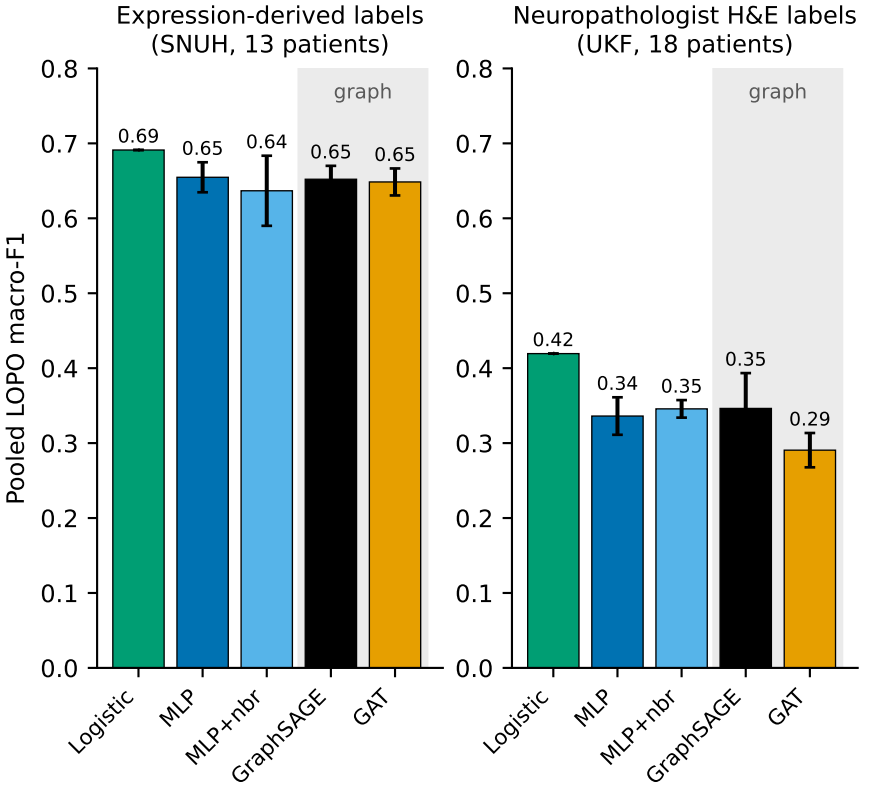
Does spatial graph structure improve supervised niche prediction in glioblastoma Visium data?



Same features, training and tuned thresholds; leave-one-patient-out
SNUH: nested tuning + ablations (1,053 runs); UKF: defaults (304 runs)



Take-home

1. Logistic regression ranked first within both cohorts (LOPO)
2. Graph encoders were never significantly better than an MLP, although homophily was 0.80
3. No depth, neighborhood size or marker-gene removal produced a graph advantage
4. Held-out patient changed macro-F1 far more than the choice of model